The University of Queensland
AUSTRALIA
CREATE CHANGE

Can AI Learn to See Like Humans?

Frequency-Based Vision for Medical Imaging

Incomplete Powerpoint currently

Youssef Hassan, Summer Research Scholar

Jevi Waugh, Summer Research Scholar

Wendi Ma, PhD Candidate

Research Supervisor: Dr. Shekhar “Shakes” Chandra

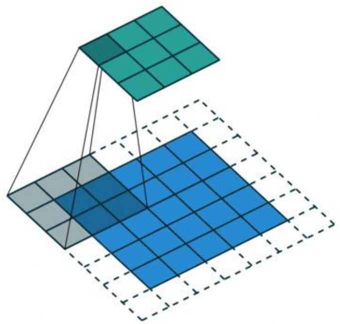
Acknowledgement of Country

- The University of Queensland (UQ) acknowledges the Traditional Owners and their custodianship of the lands on which we meet.
- We pay our respects to their Ancestors and their descendants, who continue cultural and spiritual connections to Country.
- We recognise their valuable contributions to Australian and global society.

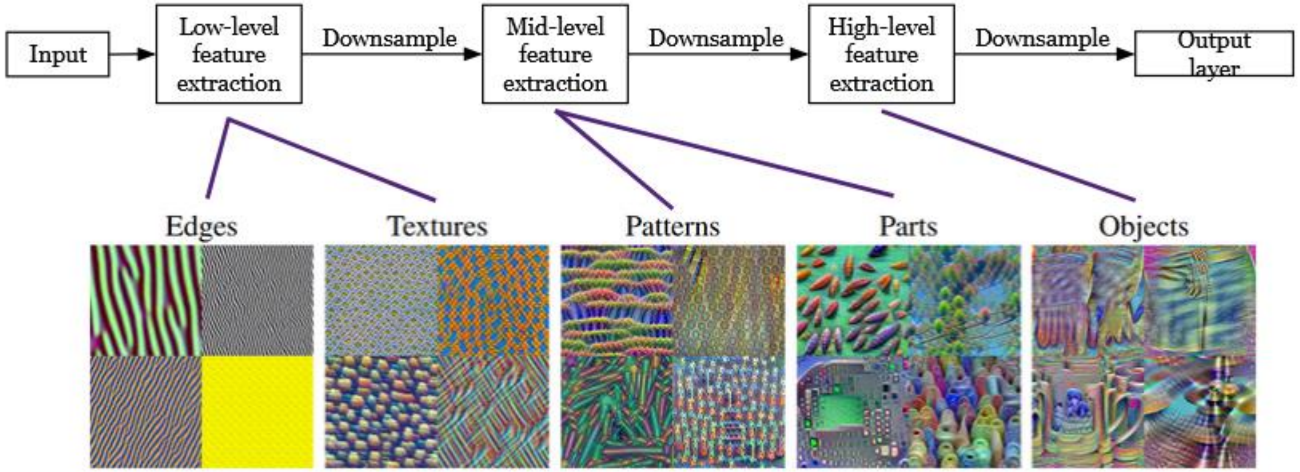
Image: Digital reproduction of *A guidance through time* by Casey Coolwell and Kyra Mancktelow

Why Rethink Vision Models?

Today's AI sees the world like this.

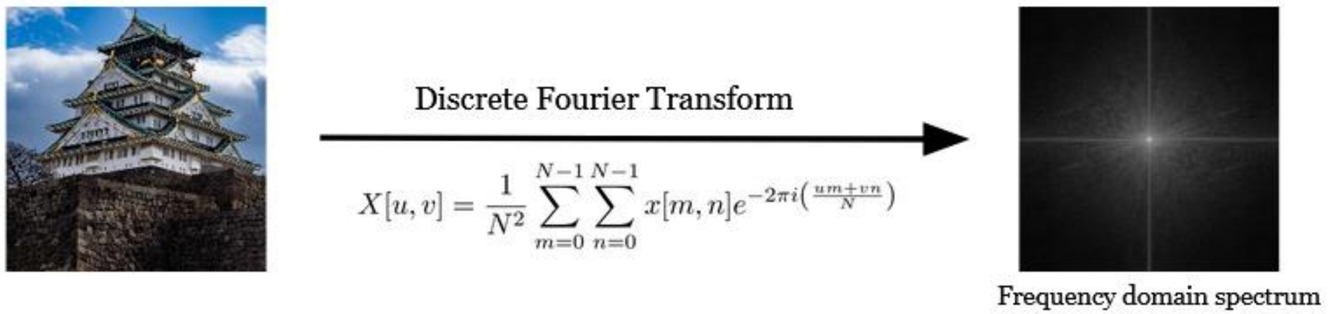


Powerful. But expensive. And hard to understand.

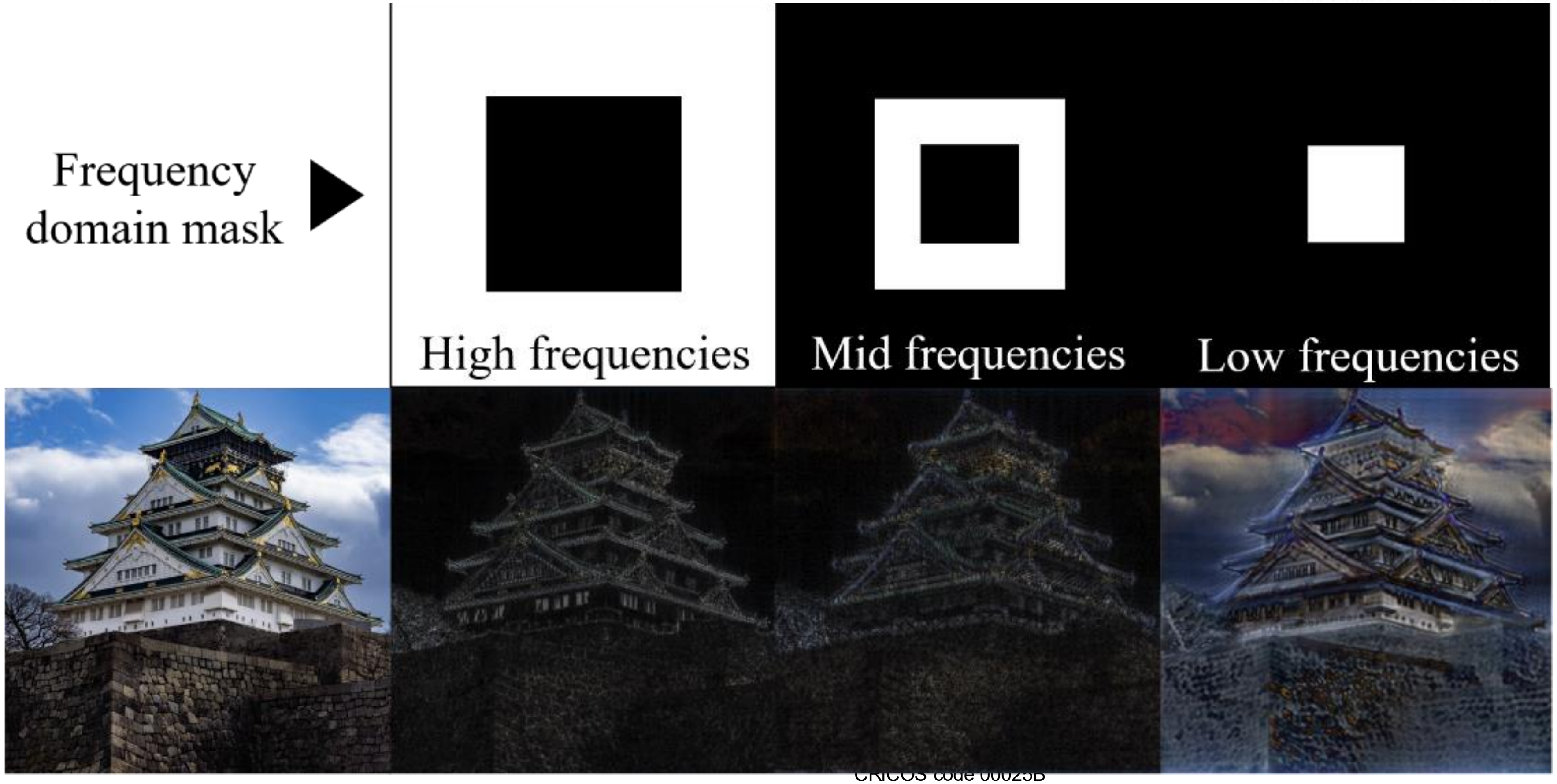


The Frequency Domain

Images are also waves!

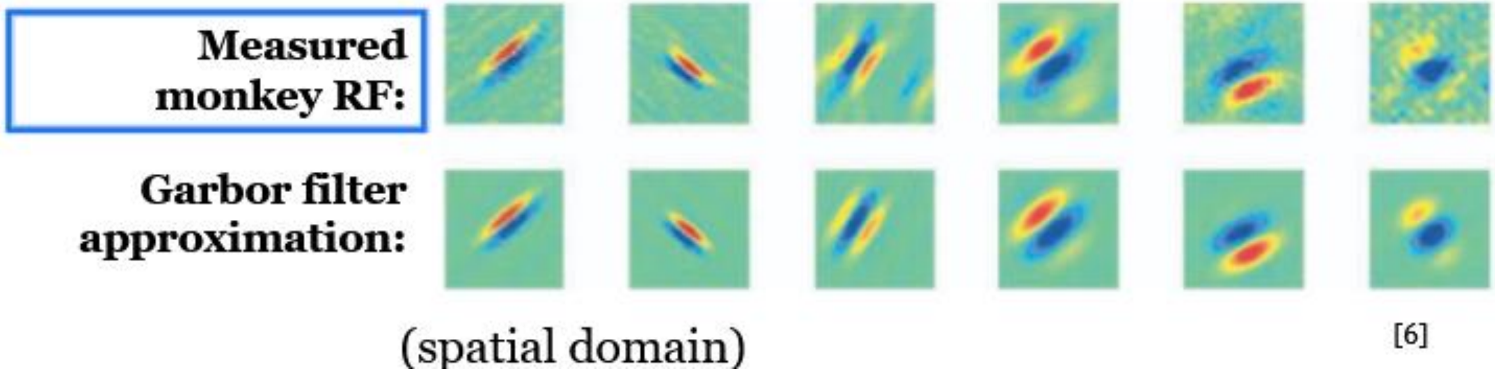


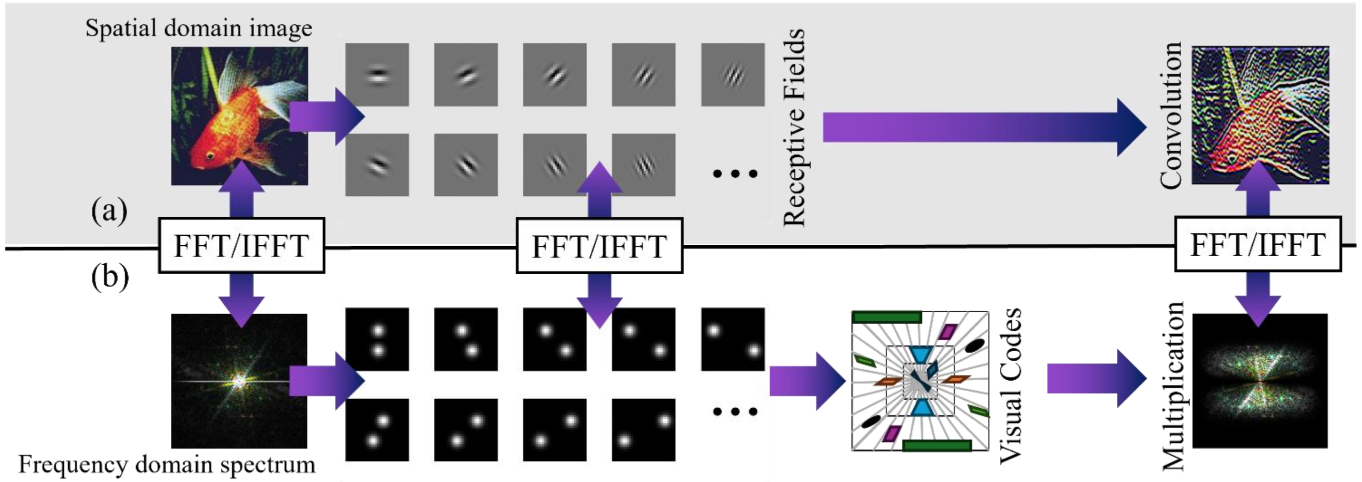
Low frequencies correspond to shapes.
High frequencies correspond to edges and fine details.

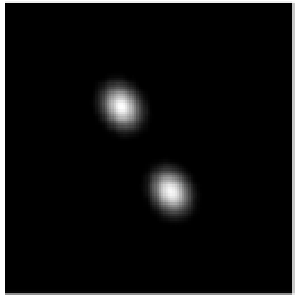


Inspiration from Human Vision

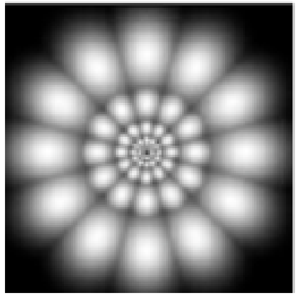
Your brain does not see pixels, it responds to frequencies!







Single Garbor filter (frequency domain)



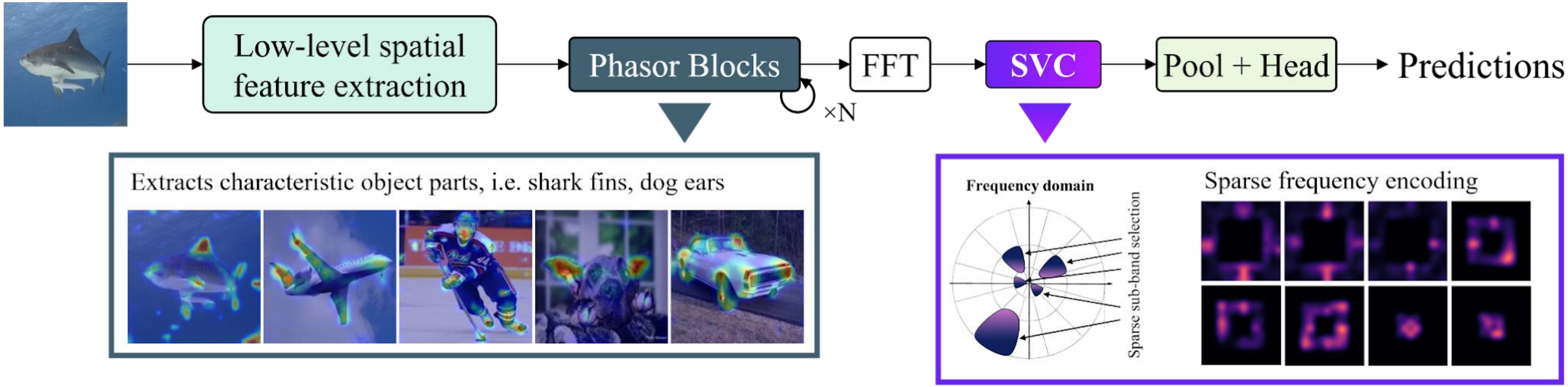
Garbor filter bank (frequency domain)

How PsychoNet Thinks Differently

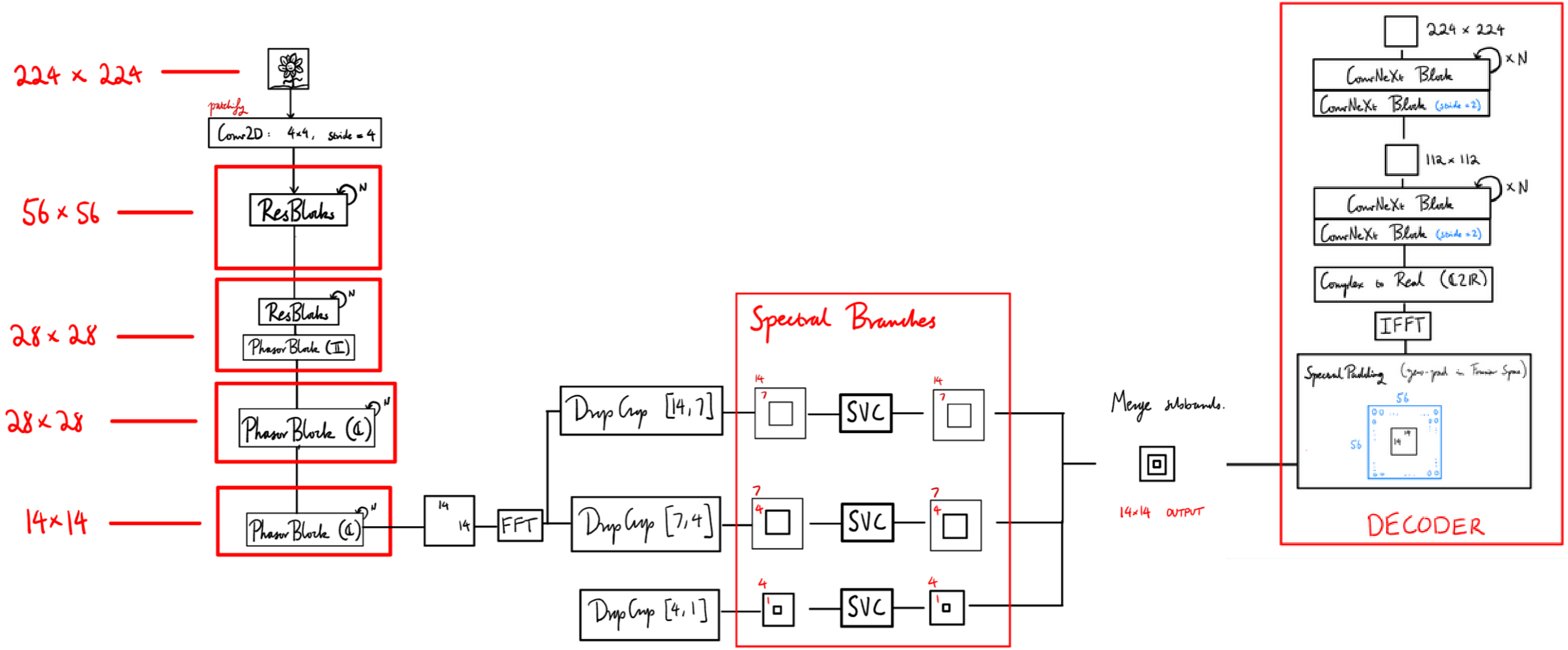
Learning What Frequencies Matter

What It Does

- Converts features into frequency space (FFT)
- Splits into radial bands (low → high)
- Learns sparse frequency codes



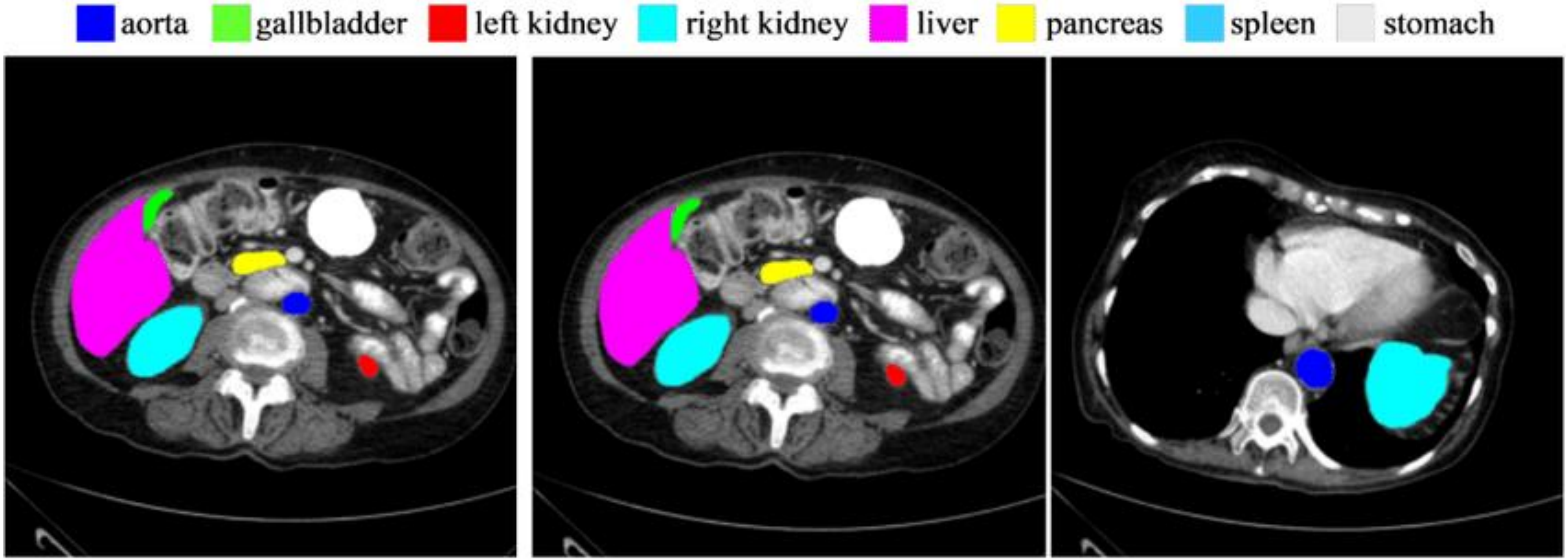
The Base PsychoNet Model



Segmentation Is Hard

Classification: What is this?

Segmentation: Where exactly is everything?



We Tried… A Lot

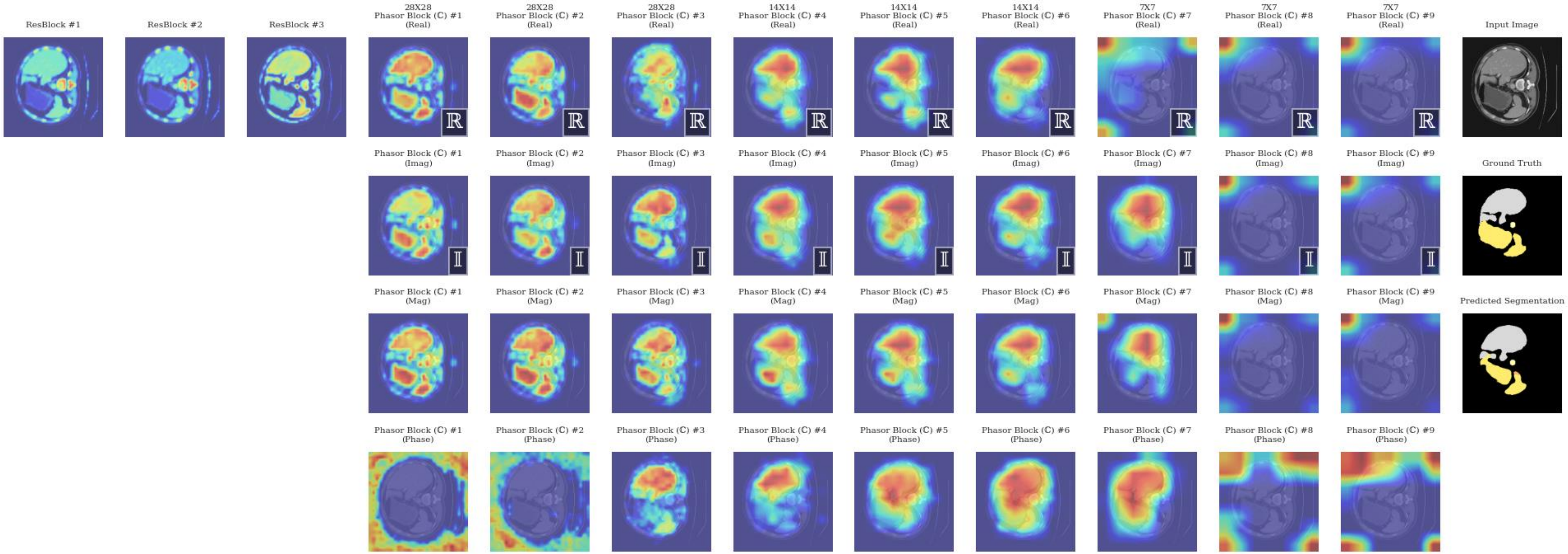
We joined a project that had been ongoing for a year and was just beginning to work on this new research angle. So, we explored many different pathways.

Decoder tweaks → Learned padding → Skip connections → Multistage backbone

But one realization changed it all…

Wait… Why Does This Already Look Like a Mask?

14x14 and 7x7 layers were redundant.



Why Were 14×14 and 7×7 Layers Redundant?

Spatial models:

Capture small → large scale iteratively

Require progressive downsampling

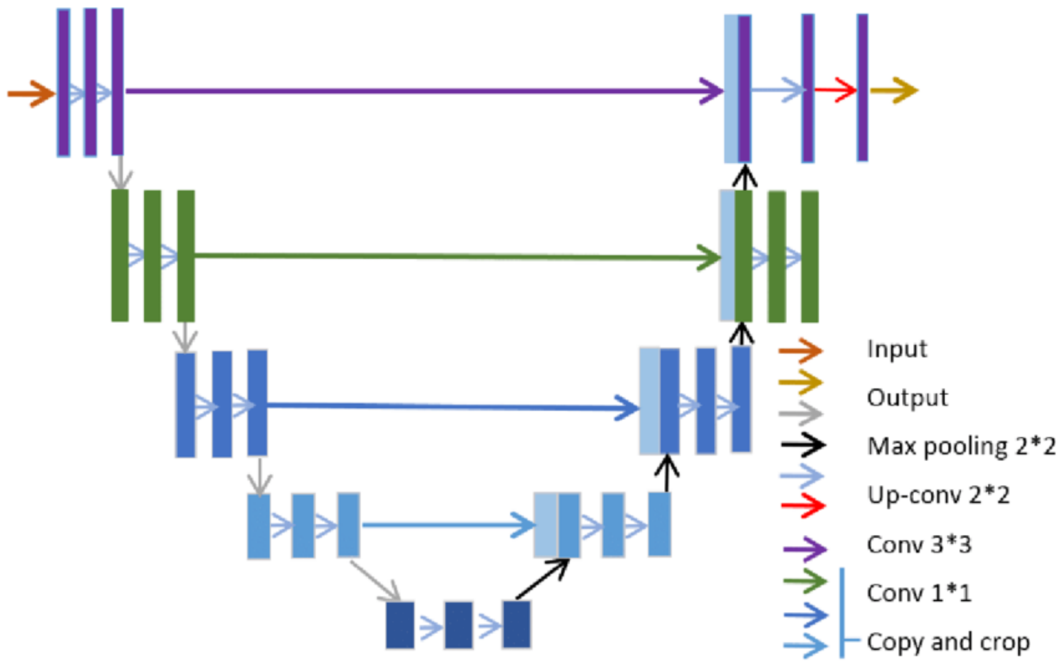
Frequency models:

Each frequency component has global support

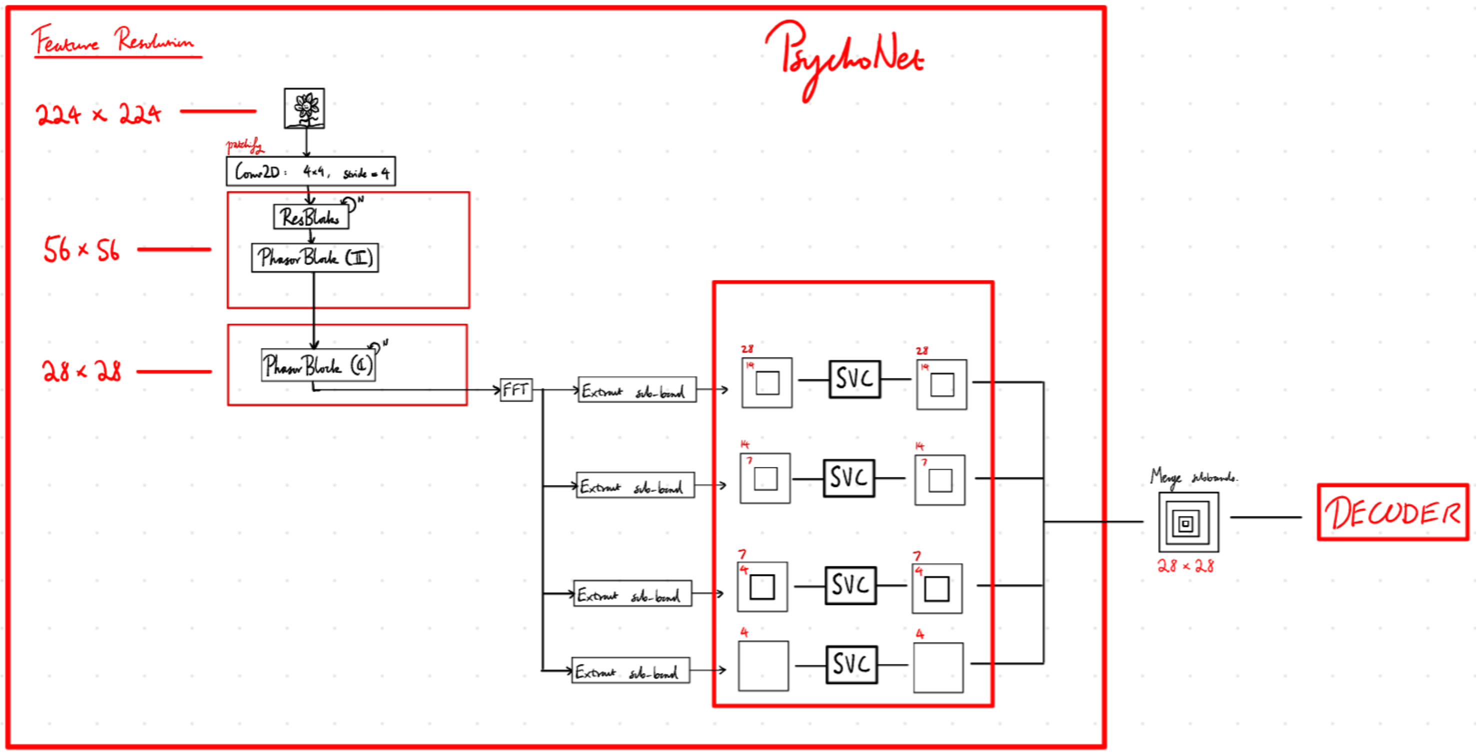
Large-scale structure captured without deep compression

Therefore:

We don’t need aggressive encoder downsampling.



So We Deleted Half the Model.



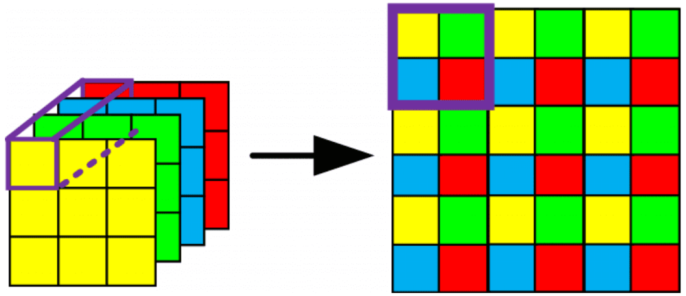
Removing Deeper Layers

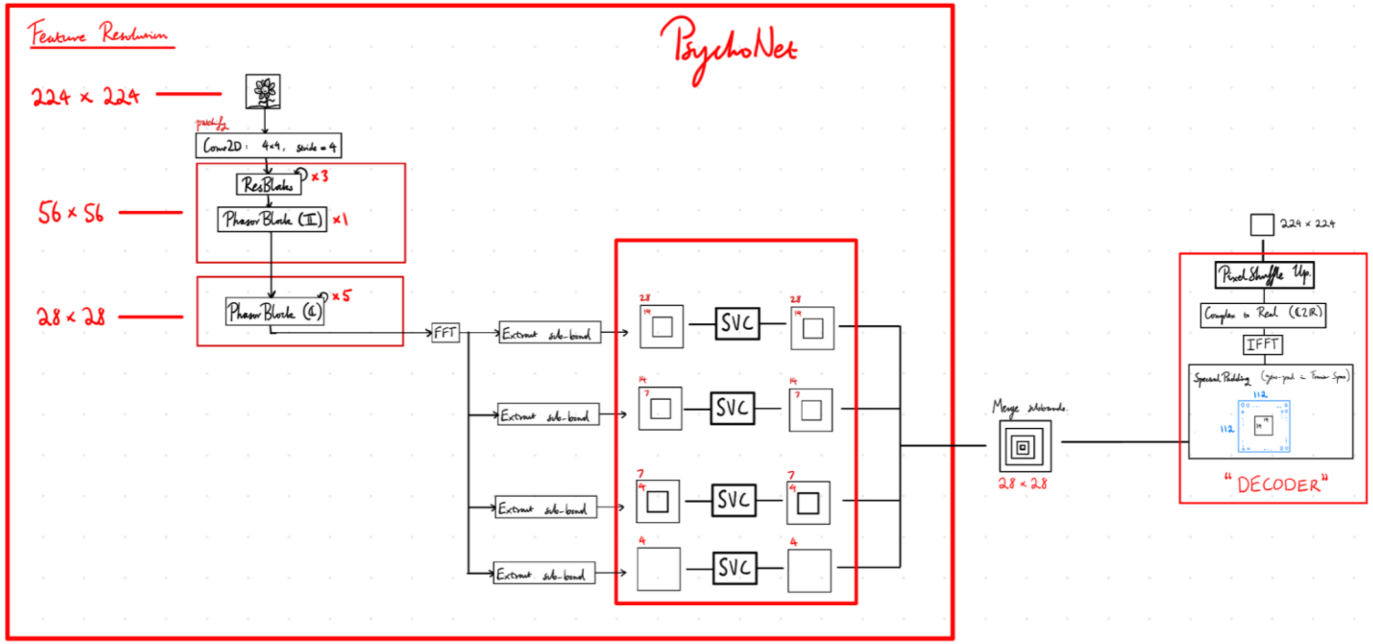
- Removed 14×14 and 7×7 stages
- Reduced parameters
- Maintained competitive performance

Do We Even Need a Big Decoder?

PixelShuffle Instead of Transpose Convolutions

- Simplified upsampling
- Removed heavy ConvNeXt decoder
- Achieved near-UNet performance
- Segmentation without a “proper” decoder.

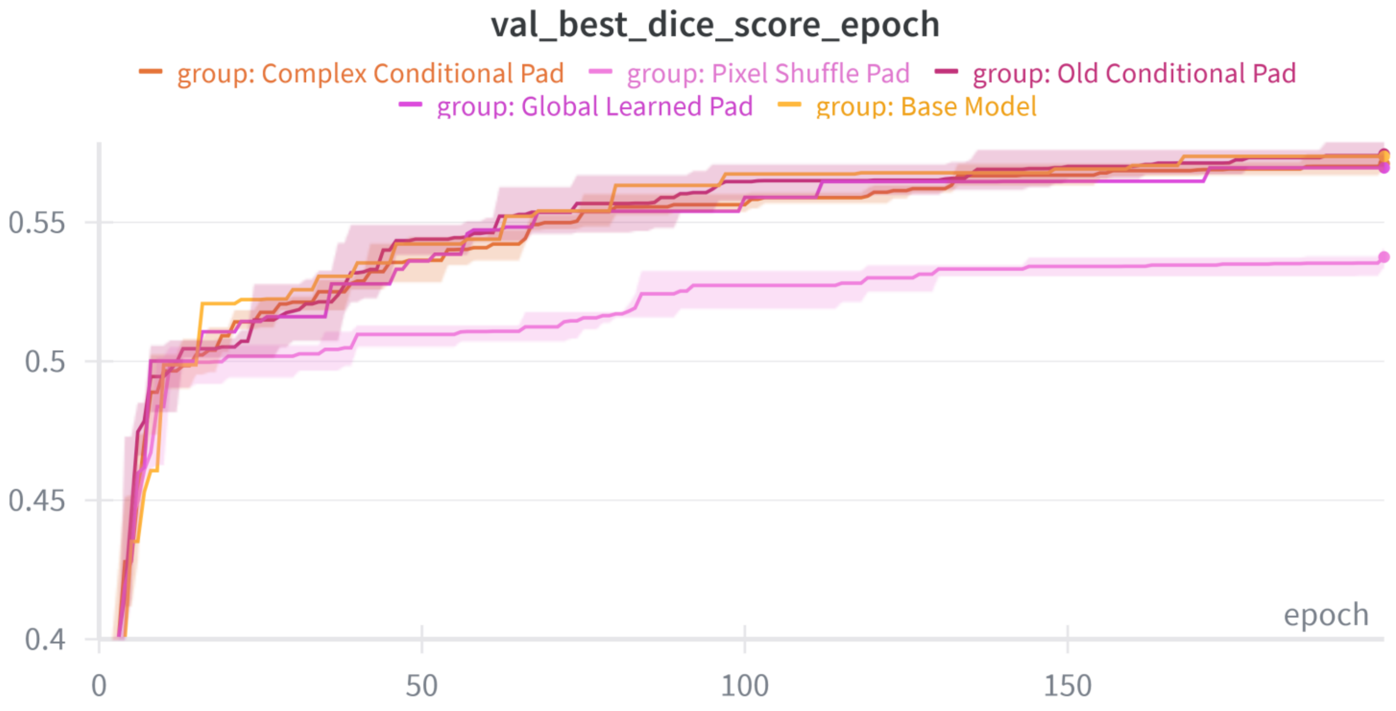




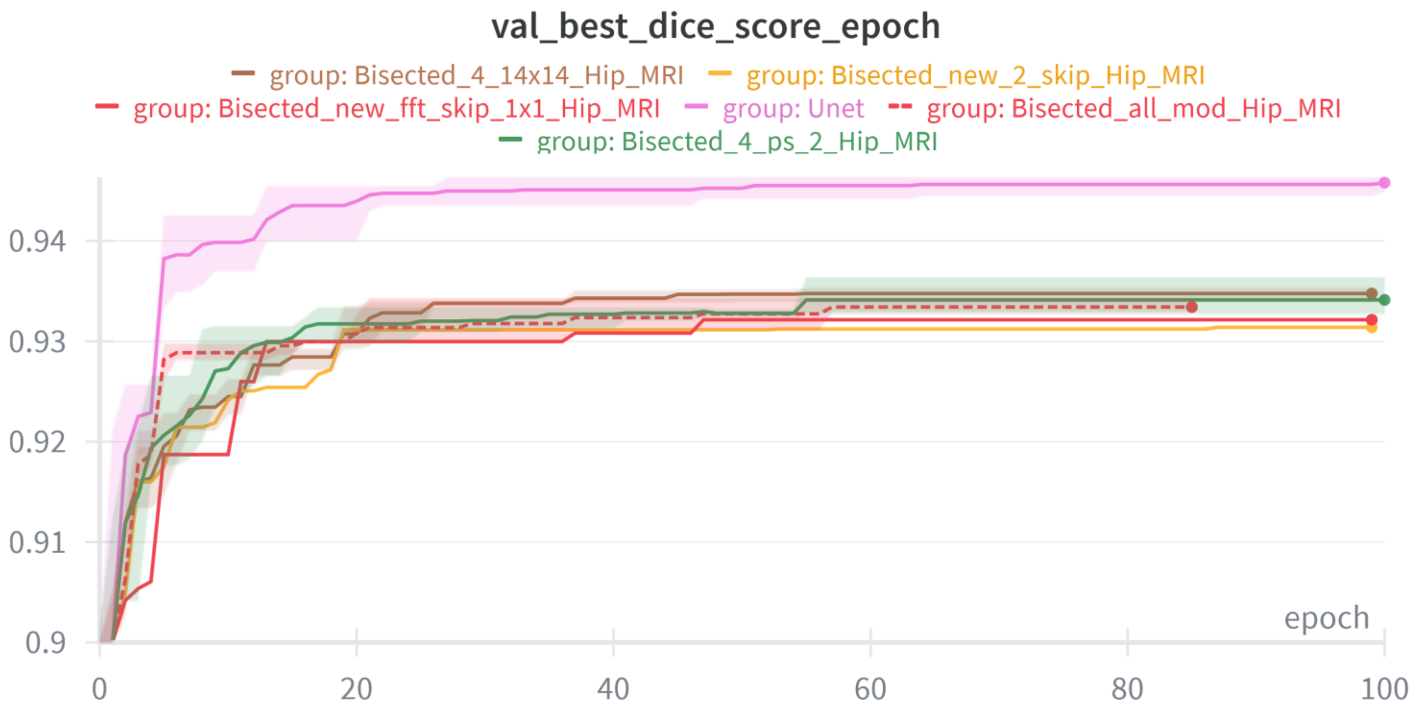
What We Learned

Key Findings

- Frequency representations can support segmentation
- Latent resolution is critical
- Decoder complexity is less important
- Sparse frequency bands remain interpretable
- Simpler models can be competitive
- Representation design can reduce architectural complexity



Testing Beyond Synapse



Hip MRI dataset:

- Larger dataset
- Harder task
- Current model does not outperform UNet

Why This Matters

Current State

Segmentation relies on deep spatial decoders.

Possible Future

Shallow frequency-based models

Greater interpretability

Biologically inspired reasoning

More efficient architectures

The Fourier Shuffle

The formula/algorithm here

Final model: Decoder that beats UNET ON Synapse

[Entity Name]

THE UNIVERSITY OF QUEENSLAND AUSTRALIA | CREATE CHANGE

Rethinking How AI Sees

Maybe AI doesn’t just need deeper networks.
Maybe it needs a different way of seeing.

Youssef Hassan
Jevi Waugh
Supervisor : Dr. Shekhar “Shakes” Chandra
Special thanks to Wendi Ma!
UQ Summer Research